

Biogas production from the anaerobic co-digestion of agro-industrial wastes: operation of a s-CSTR reactor under simulated sugarcane season and off-season conditions

Ana Beatriz Soares Aguiar ^{a*}, Renata Piacentini Rodriguez ^b, Lucas Tadeu Fuess ^c, Jens Bo Holm-Nielsen ^d, Bruna de Souza Moraes ^a

^a Center for Energy Planning (NIPE), University of Campinas. 330 Cora Coralina Street, Campinas, SP 13083-896, Brazil

^b Laboratory of Anaerobic Biotechnology, Science and Technology Institute, Federal University of Alfenas. 11,999 José Aurélio Vilela Road, Poços de Caldas, MG 37715-400, Brazil

^c Biological Processes Laboratory, São Carlos School of Engineering, University of São Paulo. 1100 João Dagnone Avenue, Santa Angelina, São Carlos, SP 15.563-120, Brazil

^d Department of Energy, Dr. Emeritus, Aalborg University. Niels Bohrs Street 6, 6700 Esbjerg, Denmark

* Corresponding author at: beatrizsoaresag@gmail.com

Center for Energy Planning (NIPE), University of Campinas. 330 Cora Coralina Street, Campinas, SP 13083-896, Brazil

Abstract

Anaerobic digestion (AD) of organic waste is a sustainable alternative for renewable energy generation and efficient waste management. In this study, the potential for biogas production from the co-digestion of vinasse, cattle manure, and slaughterhouse wastewater (blood water) in a semi-continuous stirred tank reactor (s-CSTR) was evaluated. The waste, available throughout the year, was combined in different proportions to enhance methane production and ensure system stability. The reactor operated under mesophilic conditions (35 °C) with varying organic loading rates (OLR) and substrate compositions in the feed. The highest volumetric methane production (VMP) rates were observed in co-digestion systems combining vinasse and cattle manure. A mixing ratio of 5:1 (v/m) at an OLR of 1.7 kg VS m⁻³ d⁻¹ resulted in 3,981 ± 1,207 NmL CH₄ L⁻¹ d⁻¹, whereas a 1:1 (v/m) ratio at an OLR of 4.1 kg VS m⁻³ d⁻¹ achieved a higher production of 5,269 ± 637 NmL CH₄ L⁻¹ d⁻¹. Anaerobic co-digestion of vinasse with cattle manure demonstrated high efficiency, whereas the addition of glycerol as a supplementary organic substrate compromised system stability. During the sugarcane off-season, the operation occurred by mixing blood water and cattle manure (1:2 – v/m), without adding vinasse.

The operation was successfully simulated, reaching a VMP of $3,008 \pm 1,713 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$. Energy assessment from biogas production during season and off-season periods reached 59,169 MWh annually, confirming the continuous production feasibility.

Key-words: sugarcane vinasse, cattle manure, blood water, anaerobic co-digestion, s-CSTR, energy assessment

1. INTRODUCTION

Anaerobic digestion (AD) is an efficient alternative for treating waste and effluents, enabling the recovery of methane-rich biogas and biofertilizers. This process uses microorganisms to convert organic matter into biogas, a sustainable fuel for energy generation [1].

The production of first-generation ethanol from sugarcane in Brazil is a consolidated technology, with decades of experience. This process generates organic-rich waste that can be utilized for biogas production via AD. The main wastewater from this process is vinasse, a residue with high pollutant potential due to its high concentration of organic compounds and nutrients, especially potassium, as well as nitrogen and phosphorus [2]. This byproduct is generated during the distillation of ethanol and is released at temperatures between 85–90 °C [3]. The presence of melanoidins gives it its dark brown color, while the high content of organic acids results in a low pH [4]. In Brazil, vinasse production volume varies between 10 and 15 liters per liter of ethanol [5].

For decades, vinasse has been used in sugarcane fertirrigation in Brazil. However, its high organic matter content and potential environmental impacts require attention [6]. These impacts include greenhouse gas emissions, soil salinization, metal and sulfate leaching, groundwater contamination, as well as the release of odors and attraction of insects [7–9].

An alternative approach increasingly employed in the ethanol industry is AD of vinasse, which involves biodegradation of its organic content, yielding biogas and digestate. Several studies have evaluated biogas production from this substrate [10–14]. Parsaee et al. [4] identified vinasse as a high-potential resource for biogas production, offering both economic and environmental benefits. Thus, AD has demonstrated viability as a process for biogas recovery from sugarcane vinasse. Nevertheless, operational discontinuities during the sugarcane off-season represent a limitation, compromising reactor performance and the overall energy balance of biorefineries [10,11].

Similar to the sugarcane industry, cattle farming generates biomass available throughout the year. According to IBGE, Brazil has around 238.6 million head of cattle [15], and the daily generation of

bovine manure can vary from 5 to 6% of their body weight [16]. Although traditionally used as fertilizer, cattle manure can cause significant environmental impacts when managed improperly. Direct disposal into water bodies and excessive application to soil contribute to water contamination, the emission of polluting gases, and the release of unpleasant odors [17–19]. Therefore, energy recovery through AD represents a promising alternative, particularly in co-digestion systems with substrates such as vinasse.

In addition to manure, slaughterhouses generate effluents with a high pollutant load, representing an environmental concern [20]. Among these effluents, the “red line” or “blood water” stands out, composed mainly of water used to wash the bleeding and sanitation areas containing animal blood [21].

Co-digestion of different substrates is a strategy to optimize biogas production, improving methane recovery and reducing operating costs. The combination of substrates favors the removal of organic load by correcting the C/N ratio, balancing pH, and increasing the system's buffering capacity [1,22,23]. Beyond optimizing operational parameters, this approach enables substrate integration across sectors, mitigating seasonal supply gaps such as those affecting sugarcane vinasse availability. Consequently, the proposed substrate combination offers a solution to a critical limitation in sugarcane biorefineries.

This work investigated the co-digestion of sugarcane vinasse, cattle manure, and slaughterhouse effluents for biogas production. The proposed integration may provide a viable approach for continuous biogas generation throughout the year, simultaneously mitigating environmental impacts and producing bioenergy. Process evaluation was conducted using a semi-continuous stirred-tank reactor (s-CSTR) fed with substrate mixtures at varying proportions and progressively increasing organic loading rates (OLR). These feedstocks were selected based on their year-round availability to maintain continuous operation. CSTR technology is widely adopted for biogas production, especially for high-solids substrates [24], as it offers operational flexibility to accommodate diverse feedstock characteristics, ultimately improving co-digestion efficiency and process stability.

2. MATERIAL AND METHODS

2.1 Substrate definition and feed characterization

Sugarcane vinasse, a byproduct of ethanol and sugar production, was collected during the sugarcane harvest period at Usina Ester, located in Cosmópolis, São Paulo State, Brazil. Two samples (1st and 2nd batches) of vinasse were collected, and the characterization of each was analyzed before its use in

the reactor. The cattle manure and blood water were obtained from the Nossa Senhora da Saúde slaughterhouse plant, located in Poços de Caldas, Minas Gerais State, Brazil. For inoculation, a mesophilic anaerobic inoculum was used, obtained from a full-scale CSTR, designed for biogas production from various organic wastes. The characterization of these substrates is detailed in Table 1.

Table 1. Physicochemical characteristics of the substrates evaluated: sugarcane vinasse, cattle manure, blood water, and inoculum

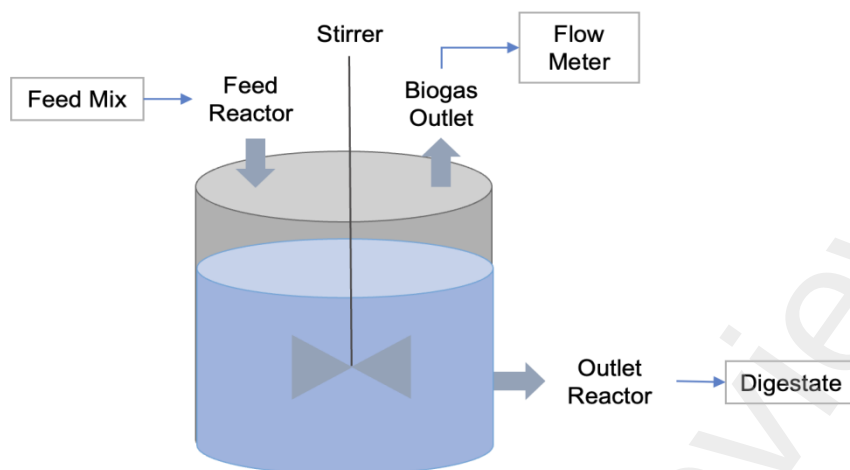
Substrate	TS (g L ⁻¹)	FS (g L ⁻¹)	VS (g L ⁻¹)	pH	COD (g L ⁻¹)
Vinasse 1st batch	34.47 ± 0.68	7.48 ± 0.21	26.99 ± 0.47	3.8	41.76 ± 0.69
Vinasse 2nd batch	24.07 ± 0.23	6.83 ± 0.06	17.24 ± 0.22	4.08	22.62 ± 0.81
Cattle Manure	181.12 ± 9.21	16.49 ± 0.21	164.63 ± 9.27	-	-
Blood Water	1.16 ± 0.02	0.29 ± 0.01	0.87 ± 0.01	7.31	1.11 ± 0.36
Inoculum	31.96 ± 0.33	13.58 ± 0.03	18.38 ± 0.33	7.9	-

TS: total solids; FS: fixed solids; VS: volatile solids; COD: chemical oxygen demand

2.2 Reactor and operating conditions

The experiment was conducted in a cylindrical anaerobic bioreactor with a conical bottom, and built-in acrylic, with a total volume of 4.9 L (Fig. 1). The reactor was equipped with stainless steel plates on the top and bottom. The top plate supported a mechanical stirrer, a feed inlet, and an outlet for biogas measurement and sampling. Gas production was continuously monitored through a flow meter (please refer to Section 2.3 for further details). At the bottom, an outlet valve connected to the reaction medium was installed.

The system's temperature was controlled using a thermostatic bath heating system connected to the reactor by hoses and equipped with a volumetric pump (Model DDC 9-7 A-PV/T/C-F-31I002FG - Grundfos®) for recirculation of the heat transfer fluid. The stirring was maintained at 240 rpm, and the operating temperature was 35 °C. The pH adjustment was performed by adding NaOH solution (0.5 mol L⁻¹) when necessary.



(a)



(b)

Fig 1. (a) Structure drawing of the semi-continuous stirred tank reactor and (b) Operational scheme on a test bench

To inoculate the biodigester, a funnel was connected to the feed tube, and 3.4 liters (working volume) of inoculum were gradually added, with periodic mixing to ensure homogeneity. Subsequently, a nitrogen (N_2) line, used as an anaerobic gas, was connected to the feed tube for approximately 15 minutes, during which the mixer was activated. After this period, the N_2 line was disconnected, the feed inlet was sealed, and the biogas outlet was opened. The digester remained under inoculation conditions for two days, after which the substrate feeding and biogas production monitoring began.

The methodology applied to inoculate and operate the s-CSTR was performed as described by Usack et al. [25].

The reactor operation was initiated with the preparation of the feed according to the previously defined hydraulic retention times (HRTs), ensuring the adaptation of the system to the established operational conditions. This preparation included the homogenization of the substrates according to the determined proportion. The main physicochemical characteristics of the reactor feed mix throughout the operational period are presented in Table 2.

During operation, the reactor was supplied with a mixture of sugarcane vinasse and cattle manure in a ratio of 5:1 (v/m). Subsequently, to increase the concentration of solids in the reactor feed, based on the operational characteristics of the s-CSTR system, the ratio of the mixture of vinasse and cattle manure was adjusted to 1:1 (v/m). Next, other mixtures between substrates were tested to evaluate the performance of blood water as a co-substrate. Additionally, glycerol (Glycerin PA ACS 1L - ACS Científica®) was introduced in select mixtures as an organic supplement due to its carbon source, highly assimilable by microorganisms under anaerobic conditions.

Table 2. Physicochemical characteristics of mixtures between substrates used as feed in the reactor

Mix	Composition	Proportion	pH	TS (g L ⁻¹)	FS (g L ⁻¹)	VS (g L ⁻¹)
Mix 1	Vinasse 1st batch + Cattle manure	5:1 (v/m)	4.41 ±0.57	50.62 ±1.82	9.18 ±0.09	41.43 ±1.88
Mix 2	Vinasse 2nd batch + Cattle manure	1:1 (v/m)	5.50 ±0.55	85.99 ±1.43	12.76 ±0.73	73.23 ±0.75
Mix 3	Blood Water + Cattle manure + Glycerol	1:1 (v/m) + 3% Glycerol*	6.78 ±0.40	98.40 ±5.16	14.04 ±0.74	84.37 ±4.42
Mix 4	Vinasse 2nd batch + Blood Water + Cattle manure	1:1:1 (v/v/m)	5.04 ±0.36	66.31 ±0.14	12.92 ±0.12	53.40 ±0.22
Mix 5	Vinasse 2nd batch + Blood water + Cattle manure + Glycerol	1:2:2 (v/v/m) + 3% Glycerol*	6.08 ±0.04	110.72 ±0.10	14.65 ±0.07	96.07 ±0.17
Mix 6	Vinasse 2nd batch + Blood water + Cattle manure	1:2:2 (v/v/m)	6.46 ±0.26	65.02 ±2.49	12.44 ±0.28	52.58±2.21
Mix 7	Blood water + Cattle manure	1:2 (v/m)	5.92 ±0.19	91.96 ±1.91	11.0 ±0.1	80.96±1.84

* Glycerol percentage refers to its volumetric proportion in the total feed mixture

The reactor was fed daily with the specific substrate blend and volume corresponding to each experimental phase. The feeding procedure involved withdrawing an aliquot of the effluent and then adding an equivalent volume of the substrate mixture to the system. In this way, the volume removed was equivalent to the volume added, ensuring the maintenance of the operating conditions characteristic of a semi-continuous regime. The experiment lasted a total of 440 days, during which different co-substrates were incorporated to simulate the conditions of the sugarcane season and off-

season periods. The parameters selected for the AD system in each experimental phase are presented in Table 3.

Table 3. Operational conditions applied to the reactor

Phase	Feed Mix	Operation period (day)	OLR (g VS L ⁻¹ d ⁻¹)	HRT (days)	Feeding rate (L d ⁻¹)	Inlet pH
I-A	Mix 1	1-70	1.4	28.6	0.12	4.12
I-B	Mix 1	71-97	1.7	24.5	0.14	4.11
I-C	Mix 1	98-145	1.2	34.3	0.10	6.05*
I-D	Mix 1	146-180	1.7	24.5	0.14	6.13*
II-A	Mix 2	181-218	1.9	38.1	0.09	5.47
II-B	Mix 2	219-285	3.0	24.5	0.14	5.63
II-C	Mix 2	286-315	4.1	18.1	0.19	5.24
III-A	Mix 3	316-322	5.2	16.3	0.21	7.33*
III-B	Mix 3	323-342	2.2	38.1	0.09	7.26*
IV	Mix 4	343-378	2.3	22.9	0.15	7.16*
V	Mix 5	379-387	2.8	34.3	0.10	6.97*
VI	Mix 6	388-412	2.3	22.9	0.15	7.06*
VII	Mix 7	413-440	3.5	22.9	0.15	6.80*

* Influent pH has been adjusted with NaOH solution

2.3 Analytical methods

System monitoring was performed based on physicochemical and operational parameters, including pH, chemical oxygen demand (COD), total volatile acids (TVA), intermediate alkalinity (IA), and partial alkalinity (PA). The analyses of pH, COD, and volatile solids (VS) followed the procedures described in the Standard Methods for the Examination of Water and Wastewater [26]. The determination of PA and IA was performed using the titration method described by Ripley et al. [27], and TVA concentrations were quantified according to the protocol established by Dilallo and Albertson [28].

The characterization of volatile fatty acids (VFAs), alcohols, and sugars was performed by High-Performance Liquid Chromatography (HPLC) (PROMINENCE HIC-SP/HIC-NS, Shimadzu®), using a set equipped with an automatic liquid sampler (SIL-20A) and a UV-PDA detector set at 210 nm for organic acids. Before analysis, samples were filtered through 0.22 µm syringe filters and acidified with 40 µL of 1N H₂SO₄ in vials containing 1 mL of the sample [29].

Biogas flow rates were monitored by a gas meter (MilliGascounter MGC-1 PMMA, Dr.-Ing. Ritter Apparatebau GmbH & Co. KG, Germany) connected to the reactor headspace. The biogas composition, including methane (CH₄), carbon dioxide (CO₂), hydrogen sulfide (H₂S), and hydrogen (H₂), was analyzed using gas chromatography (Nexis GC-2030, Shimadzu®).

2.4 Equations used in response variable calculation

The performance of the s-CSTR was evaluated using the following response variables: effluent and influent pH, effluent and influent VS (g SV L^{-1}), intermediate alkalinity-to-partial alkalinity ratio (IA/PA), biogas flow rate (NmL d^{-1}), biogas composition ($\%\text{CH}_4$, $\%\text{CO}_2$, $\%\text{H}_2\text{S}$ and $\%\text{H}_2$), methane yield (MY) ($\text{NmL CH}_4 \text{ g}^{-1} \text{ VS added}$), and VMP ($\text{NmL CH}_4 \text{ L}^{-1} \text{ d}^{-1}$). The equations used to calculate these response variables are provided below (Equations 1–3).

The values reported for biogas production-related response variables correspond to standard conditions of normal temperature and pressure (STP), defined as 1 atm and 0°C (273.15 K).

$$\text{Biogas flow} = \frac{V_{\text{Biogas production}}}{\text{time}} \quad (1)$$

The *Biogas flow* is expressed in mL d^{-1} and $V_{\text{Biogas production}}$ refers to the total volume of biogas produced over a given time (day) period and directly measured in the gas meter.

$$\text{Volumetric methane production (VMP)} = \frac{\text{Biogas flow} \times \%\text{CH}_4}{\text{Useful volume}_{\text{reactor}}} \quad (2)$$

The volumetric methane production (VMP) is expressed in $\text{mL CH}_4 \text{ L}^{-1} \text{ d}^{-1}$, $\%\text{CH}_4$ is the methane content in the biogas (percentage), and $\text{Useful volume}_{\text{reactor}}$ is the effective volume of the reactor, given in liters.

$$\text{Methane yield (MY)} = \frac{\text{Biogas flow} \times \%\text{CH}_4}{Q_{\text{In}} \times \text{VS}_{\text{add}}} \quad (3)$$

The methane yield (MY) is expressed in $\text{mL CH}_4 \text{ g}^{-1} \text{ VS added}$, Q_{In} refers to the influent flow of the system, given in L d^{-1} and VS_{add} is the VS concentration added to the system, given in g VS L^{-1} .

2.5 Energy potential estimation

The energy potential of the system was estimated using Equations 4 and 5. *Methane potential* is expressed in $\text{mL CH}_4 \text{ L}^{-1}$ and *Energy potential* in MJ m^{-3} .

$$\text{Methane potential} = \text{MY} \times \text{VS} \quad (4)$$

$$\text{Energy potential} = \text{Methane potential} \times \text{LHV}_{\text{CH}_4} \quad (5)$$

LHV_{CH_4} is the lower heating value of CH_4 , considered as $35.8 \text{ MJ m}^{-3} \text{ CH}_4$.

For the energy potential assessment, the baseline parameter was set at 1 million tons of sugarcane crushing per season. Based on typical mass balances of attached mills, vinasse production is

approximately 450 liters per ton of crushed cane [30], yielding 450 million liters during the harvest period for the evaluated scenario.

Two operational scenarios were defined for the reactor over an annual cycle, covering the sugarcane harvest and off-season periods. Scenario 1 employed Mix 1 (Phase I-D) during harvest and Mix 7 (Phase VII) during the off-season. The harvest period (7.5 months) involved treating 450,000 m³ of vinasse co-digested with cattle manure at a 5:1 ratio (v/m). Under these conditions, 90,000 tons of cattle manure (~12,000 tons/month) were required, corresponding to the daily production of ~15,000 cattle at an average rate of 27.8 kg animal⁻¹ day⁻¹ [31].

During the off-season (4.5 months), the system was fed with Mix 7, composed of blood water and cattle manure at a 1:2 ratio (v/m). To maintain continuous operation and optimize installed infrastructure utilization, the reactor's operating capacity was kept constant across both periods, avoiding the idle phase commonly observed in biogas plants during substrate shortages. Maintaining the same operating volume required tripling the monthly manure supply rate during the off-season to ~36,000 tons/month, supplemented with ~18,000 m³/month of blood water as co-substrate.

Scenario 2 employed Mix 2 (Phase II-B) during harvest, with a 1:1 vinasse-to-cattle manure ratio (v/m). Treating the total vinasse volume (450,000 m³) would require an equivalent manure input of 450,000 tons (~60,000 tons/month) from a herd of ~72,000 head. However, this configuration posed significant logistical challenges and would substantially increase the required substrate processing volume, compromising technical and economic viability. Consequently, only 50% of the vinasse volume (225,000 m³) was considered for treatment, requiring 225,000 tons of manure (~30,000 tons/month) from ~25,000 animals, a more operationally realistic configuration compatible with substrate supply capacity.

During the off-season, Mix 7 was maintained to ensure full utilization of the reactor's operating volume. This phase required monthly waste supply rates 1.2 times higher than during harvest (~36,000 tons/month of manure), supplemented with ~18,000 m³/month of blood water as co-substrate.

3. RESULTS AND DISCUSSION

3.1 Operational parameters and system control

Figure 2 illustrates the operational performance of the reactor, highlighting variations in pH, alkalinity, and organic matter throughout the different operating phases. The system operation started with Mix

1, composed of vinasse and cattle manure in a ratio of 5:1 (v/m) (Table 2). This phase lasted for 180 days, with the application of different OLRs. Initially, in Phase I-A, the OLR adopted was $1.4 \text{ kg VS m}^{-3} \text{ d}^{-1}$, without the addition of alkalizing agents. The mean inlet pH was 4.12 ± 0.04 , while the mean outlet pH reached 7.17 ± 0.16 , a value considered adequate for anaerobic digestion [24,32]. The mean IA/PA ratio observed during Phase I-A was 1.31 ± 0.52 . By the end of the phase, the mean value calculated for the final three measurements had decreased to 0.74 ± 0.03 , already indicating signs of system stabilization. Furthermore, a progressive reduction in TVA concentration was observed throughout the period, decreasing from $9,630 \text{ mg L}^{-1}$ on day 8 (the highest value recorded during this phase) to $4,545 \text{ mg L}^{-1}$ on day 69 (the final of this phase) (Fig. 1b).

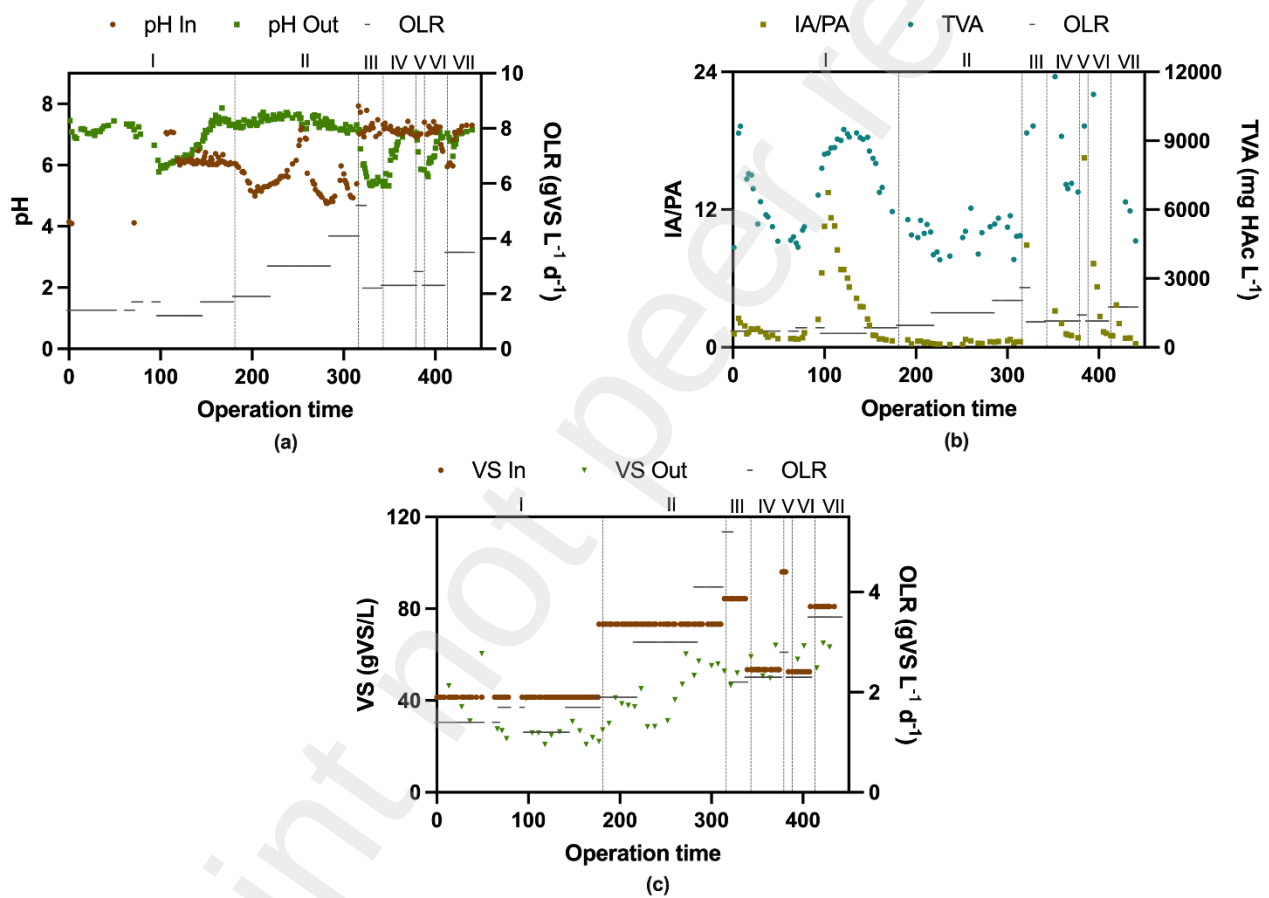


Fig 2. Temporal profiles for the (a) inlet and outlet pH, (b) IA/PA ratio and TVA concentration, (c) inlet and outlet VS concentration

*The vertical dashed line indicates the division of operating phases

Subsequently, Phase I-B was characterized by an increase in OLR to $1.7 \text{ kg VS m}^{-3} \text{ d}^{-1}$, maintaining the absence of alkalizing agents. The mean inlet pH remained close to that of the previous phase (4.11), while the mean outlet pH decreased to 6.86 ± 0.40 . In this stage, the IA/PA ratio showed a mean of 2.34 ± 2.42 , evidencing signs of operational instability. Kunz et al. [33] suggested an optimal IA/PA

ratio between 0.3 and 0.4, however, some variations may occur due to the characteristics of the reactor and the substrate. In addition, the decrease in effluent pH and the increase in TVA concentration indicated unfavorable conditions for methanogenic activity, thereby compromising the efficiency of the anaerobic digestion process. Given this scenario, it was decided to reduce the OLR again, aiming at recovering operational stability.

The OLR was adjusted to $1.2 \text{ kg VS m}^{-3} \text{ d}^{-1}$ in Phase I-C and the inlet pH correction was started by dosing NaOH. With this measure, the mean inlet pH increased to 6.05 ± 0.76 , resulting in a mean outlet pH of 6.23 ± 0.26 . The IA/PA ratio, although still high around 7.54 ± 3.30 , showed a reduction throughout the phase, varying from 10.54 to 3.52. Given this, it was decided to increase the OLR again to $1.7 \text{ kg VS m}^{-3} \text{ d}^{-1}$, starting Phase I-D.

In Phase I-D, the influent pH correction was maintained, reaching a mean of 6.13 ± 0.12 . As a result, the mean effluent pH increased to 7.33 ± 0.24 , and the IA/PA ratio reduced markedly, reaching a mean of 1.14 ± 0.68 . These results indicate that the system showed signs of operational recovery, creating more favorable conditions for anaerobic digestion and the consequent biogas production.

During Phase I, which corresponded to the system's adaptation period, signs of organic matter removal were observed, as indicated by a reduction in VS concentration in the effluent, meaning $35.8 \pm 10.8\%$. This trend reflects the onset of organic matter degradation. However, given the inherent instability of the startup phase, removal efficiency was comparatively lower. Sporadic increases in VS concentration were also recorded (Fig. 2c), which may be attributed to samples containing microbial biomass, potentially resulting from active microbial growth and/or the release of cellular material within the reactor.

Phase II was mainly characterized by an increase in the concentration of VS in the feed, since the properties of a mixing reactor allow operation with substrates or mixtures containing a higher concentration of solids [24] up to approximately 15% [34]. Thus, Mix 2 was prepared with vinasse and cattle manure in a 1:1 ratio (v/m) (Table 2), resulting in an increase in the concentration of total solids from 5% to 9%. Phase II-A started on day 181, with an OLR of $1.9 \text{ kg VS m}^{-3} \text{ d}^{-1}$, without correction of the influent pH, since Mix 2 presented a higher pH than Mix 1, around 5.5 ± 0.55 . Thus, during Phase II, there was no need to adjust the influent pH through the addition of alkalizing agents. In Phase II-A, the mean inlet pH was 5.47 ± 0.34 , resulting in a mean outlet pH of 7.39 ± 0.14 and a mean IA/PA ratio of 0.45 ± 0.12 . In Phase II-B, the OLR was increased to $3.0 \text{ kg VS m}^{-3} \text{ d}^{-1}$ in order to optimize the system performance. At this stage, the mean outlet pH was 7.49 ± 0.15 , while the mean IA/PA ratio was reduced to 0.38 ± 0.15 . In Phase II-C, the OLR was increased to $4.1 \text{ kg VS m}^{-3} \text{ d}^{-1}$,

maintaining the operation without the use of alkalizing agents. This phase was marked by a slight reduction in the mean effluent pH (7.21 ± 0.10), along with increases in the mean IA/PA ratio (0.51 ± 0.10) and in the mean TVA concentration ($5,068 \pm 647$ mg HAc L⁻¹). Nevertheless, the system remained operational, sustaining stable biogas production levels, as discussed in Section 3.2. Furthermore, the mean VS removal observed during Phase II was $43.1 \pm 14.7\%$, indicating an improvement in organic matter removal efficiency.

The technological limitations associated with anaerobic processes are primarily related to the start-up phase and the maintenance of active biomass within the reactor [35]. In certain situations, measures are taken to prevent the forced shutdown of the treatment system to preserve the continuity of energy production. In this context, Phase III aimed to simulate an off-season period in which vinasse is not available. This strategy aimed to ensure the continuous operation of the reactor, thereby preventing shutdowns that could pose technical challenges and limit the energy balance of sugarcane distilleries [10,35]. Thus, the system was fed with Mix 3, composed of blood water and cattle manure in a ratio of 1:1 (v/m), with the addition of 3% glycerol (Table 2). The introduction of glycerol was intended to compensate for the absence of organic matter from the vinasse, since the blood water has distinct properties and a lower concentration of organic matter. In addition, glycerol, which is a byproduct of yeast metabolism under osmotic stress [36], is one of the primary organic compounds found in vinasse [37,38], characterizing a suitable substrate for the anaerobic consortium established in the s-CSTR.

In Phase III-A, the OLR was increased to 5.2 kg VS m⁻³ d⁻¹, without correction of the influent pH, since Mix 3 presented an initial pH of 7.33 ± 0.46 . However, the results were unsatisfactory, since the substrate composition resulted in inhibitory effects on the anaerobic digestion process. This was reflected in the decrease of the outlet pH (6.56 ± 0.55), in the marked increase of the IA/PA ratio (8.90), and in the elevation of the TVA concentration to $9,345$ mg HAc L⁻¹. These parameters indicated an imminent collapse of the system. Given this scenario, the OLR was reduced to 2.2 kg VS m⁻³ d⁻¹ (Phase III-B), to mitigate the reactor overload. However, the results obtained continued to show signs of instability and a drop in process efficiency, evidenced by a new reduction in the outlet pH (5.51 ± 0.18) and by the additional increase in the TVA concentration ($9,645$ mg HAc L⁻¹).

Several studies highlight glycerol as an efficient co-substrate for AD [14,39–42]; however, its proportion within the substrate mixture is a critical factor influencing process performance. Its comparative advantages include high anaerobic biodegradability; its liquid nature, which facilitates implementation through simpler feeding equipment; and its high organic content, which enables an increase in the OLR with minimal impact on the HRT [43]. Holm-Nielsen et al. [44] report that the addition of glycerol can enhance biogas yields, provided its concentration within the fermenter does

not exceed the limit of 5–7 g L⁻¹. Beyond this limit, further increases may lead to an imbalance in the AD process. The authors observed that the addition of glycerol caused an organic overload in the system, resulting in slow degradation of organic matter and, consequently, accumulation of substrate in the reactor. Consequently, VFA concentrations increased in the fermenter, suggesting that the overload was due to inhibition of the methanogenic stage [44]. The inhibition of AD by glycerol occurs in some cases because it is a compound readily available for use by fermentative bacteria; thus, the limiting step may be the acetogenic or methanogenic stage. In this case, some organic acids (acetic, propionic, butyric, valeric, and others) are formed by fermentative bacteria and may not be consumed by acetogenic bacteria at the same rate at which they are produced, further compromising methanogens [42].

Ferreira et al. [41] investigated the effect of adding crude and pure glycerol in co-digestion with sewage sludge, testing mixtures with sludge-to-glycerol ratios of 100:0, 95:5, 90:10, 85:15, and 80:20 (% v/v). In all combinations containing glycerol, a pronounced pH drop was observed, resulting in process inhibition. The authors attributed this acidification to the high biodegradability of glycerol, which led to the accumulation of volatile fatty acids. The accumulation of these compounds can compromise the stability of the anaerobic co-digestion process. Goembira et al. [45] evaluated the technical feasibility of biogas production through the addition of crude glycerol derived from a palm oil processing plant, using anaerobic reactors fed with a 1:1 mixture of cattle manure and distilled water. Glycerol concentrations of 5% (GL5), 10% (GL10), and 15% (GL15) were tested. The highest biogas yield was observed in GL10 (380 mL g⁻¹ VS). Conversely, the addition of 15% crude glycerol led to a complete inhibition of biogas production. The authors attributed this outcome to the excessive glycerol concentration, which may cause rapid shifts in pH, hindering the adaptation of methanogenic microorganisms and compromising overall process stability. Thus, as noted, organic overload and subsequent inhibition are often reported as the main risks to process stability during glycerol co-digestion.

In Phase IV, the main objective was to promote system recovery by removing glycerol from the feed and reintroducing vinasse into the process. The reactor was thus supplied with Mix 4, consisting of vinasse, blood water, and cattle manure in a 1:1:1 (v/v/m) ratio (Table 2), while maintaining the same OLR as in the previous phase, at 2.3 kg VS m⁻³ d⁻¹. Given that the initial pH of the mixture was 5.04, the influent pH correction was performed to create more favorable conditions for anaerobic digestion, resulting in an adjusted pH of 7.16 ± 0.12.

During this phase, which lasted 36 days, the system showed signs of operational recovery. The mean outlet pH increased progressively, reaching 6.44 ± 0.44. In addition, there was a reduction in the mean

IA/PA ratio (1.56 ± 0.90) and in the TVA concentration, which decreased to $6,765 \text{ mg HAc L}^{-1}$ at the end of Phase IV. Although the system was not yet operating at its maximum efficiency, the results indicated a positive recovery trend, suggesting a possible stabilization of the process.

Phase V aimed to continue simulating the gradual transition from vinasse to blood water by further reducing the vinasse content in the feed mixture. Accordingly, the effect of glycerol addition was reassessed, this time in the presence of a small proportion of vinasse. The system was fed with a mixture consisting of vinasse, blood water, and cattle manure in a 1:2:2 (v/v/m) ratio, supplemented with 3% glycerol (Table 2). However, as anticipated, the system again exhibited signs of collapse, with a mean effluent pH of 6.54 ± 0.52 , an IA/PA ratio of 16.5, and a TVA concentration of $9,645 \text{ mg HAc L}^{-1}$. In light of these results and the evident operational instability, Phase V was concluded after nine days, marking the beginning of Phase VI.

In Phase VI, the reactor was fed with Mix 6, composed of vinasse, blood water, and bovine manure, in the proportion 1:2:2 (v/v/m) (Table 2), this time without the addition of glycerol. The applied OLR was $2.3 \text{ kg VS m}^{-3} \text{ d}^{-1}$, a value lower than that of the previous phase due to the instability previously observed in the system. The inlet pH was corrected to 7.06 ± 0.29 . Under these conditions, the reactor presented a mean outlet pH of 6.36 ± 0.47 , a mean IA/PA ratio of 3.56 ± 2.65 , and a TVA concentration of $11,025 \text{ mg HAc L}^{-1}$. Despite the high concentration of volatile acids resulting from the instability observed in the previous phase, a reduction in the IA/PA ratio was noted, decreasing from 7.29 to 1.21. This behavior suggests that the system was showing signs of recovery, with an increase in buffering capacity and generation of alkalinity throughout the phase.

In Phase VII, the main objective was to simulate the reactor operation without the addition of vinasse, representing the off-season period. For this stage, the system feed consisted of a mixture of blood water and cattle manure in a ratio of 1:2 (v/m) (Table 2). Considering the reactor stabilization in the previous phases, the OLR was increased to $3.5 \text{ kg VS m}^{-3} \text{ d}^{-1}$, aiming to optimize system performance. The influent pH was corrected to 6.80 ± 0.53 , ensuring more favorable conditions for anaerobic digestion. During this phase, the operational parameters indicated an improvement in the system performance. An increase in the mean outlet pH to 6.88 ± 0.25 , a reduction in the IA/PA ratio, which reached 1.38 ± 1.15 , and a decrease in the TVA concentration to $5,635 \pm 888 \text{ mg HAc L}^{-1}$ were observed. These results indicate that the reactor adapted to the new operational conditions, achieving adequate performance and enabling continuous operation during the off-season in the absence of vinasse.

3.2 Methane production

Figure 3 presents the results of biogas composition and methane production in each experimental phase. During the initial phase of reactor operation (Phase I-A), the feed was carried out with Mix 1, composed of vinasse and cattle manure (5:1, v/m) (Table 2), under an OLR of $1.4 \text{ kg VS m}^{-3} \text{ d}^{-1}$. Under these conditions, a mean methane flow rate of $736 \pm 316 \text{ NmL CH}_4 \text{ d}^{-1}$, MY of $148 \pm 64 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS feed}$, VMP of $2,501 \pm 1,076 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$, and a mean CH_4 concentration of $53 \pm 11\%$ were obtained. In Phase I-B, the OLR was increased to $1.7 \text{ kg VS m}^{-3} \text{ d}^{-1}$, maintaining the absence of alkalizing agents. The combination of the increased organic load and low feed pH contributed to operational instability in the system, evidenced by the reduction in the effluent pH, together with the increase in the IA/PA ratio and the TVA concentration, as described in Section 3.1. This imbalance affected process performance, which was reflected in the methane production. Between days 71 and 97, corresponding to the beginning and end of this phase, the methane content in the biogas declined from 56.2% to 35.6% (Fig. 3c). Concurrently, the MY decreased from 123 to 66 $\text{NmL CH}_4 \text{ g}^{-1} \text{ VS feed}$, and the VMP dropped from 2,420 to 1,300 $\text{NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$ (Fig. 3b).

As a corrective measure, in Phase I-C, the OLR was reduced to $1.2 \text{ kg VS m}^{-3} \text{ d}^{-1}$, and the feed pH was adjusted to approximately 6.05 ± 0.76 . This strategy appears to have contributed to the stabilization and gradual recovery of the system. Subsequently, in Phase I-D, the OLR was increased to $1.7 \text{ kg VS m}^{-3} \text{ d}^{-1}$, maintaining the inlet pH correction (6.13 ± 0.12). As a result, there was an increase in methanogenic productivity, reaching a mean methane flow rate of $1,171 \pm 355 \text{ NmL CH}_4 \text{ d}^{-1}$, an MY of $202 \pm 61 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS feed}$, and a VMP of $3,981 \pm 1,207 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$, with an increase in the mean CH_4 concentration to $58.4 \pm 6.3\%$. These results indicate that, under the given feed mix and experimental conditions, OLRs exceeding $1.4 \text{ kg VS m}^{-3} \text{ d}^{-1}$ necessitate pH adjustment of the feeding flow to ensure adequate buffering capacity of the system. Considering scale-up opportunities, utilizing low-cost alkalizing agents is *sine qua non* to achieve economic suitability in vinasse anaerobic digestion [38], so defining a fine tune between the applied OLR (the higher the OLR, the lower the investment cost) and chemical dosing (the higher the dose, the higher the operating cost) demands proper investigation.

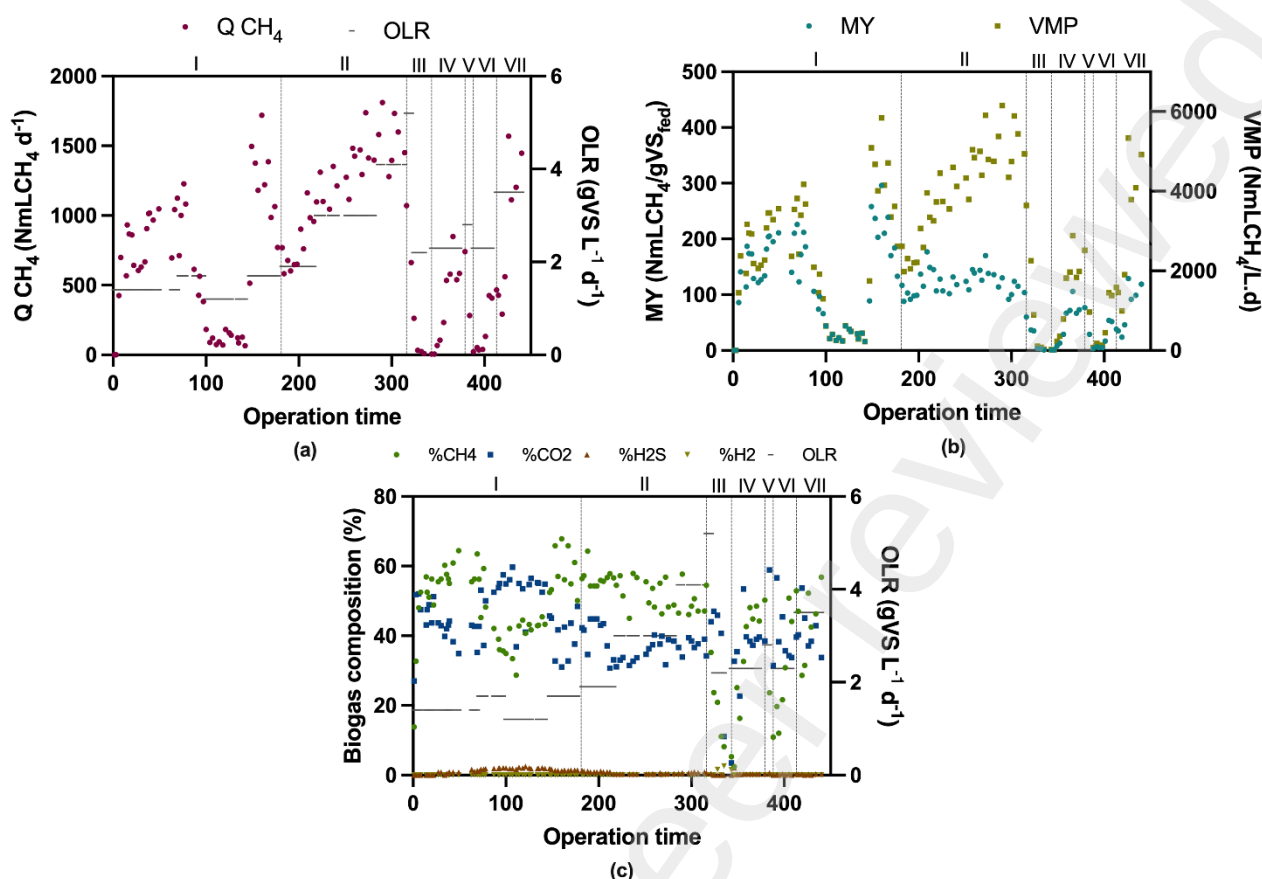


Fig. 3 Temporal profiles for the (a) daily methane flow (Q_{CH_4}), (b) methane yield (MY), and volumetric methane production (VMP), and (c) biogas composition

In Phase II, the vinasse-to-cattle manure ratio was adjusted to 1:1 (v/m) (Table 2). This modification led to an increase in total solids concentration and elevated the influent pH to approximately 5.50 ± 0.55 , attributable to the higher manure proportion compared to Phase I. Given this scenario, it was decided to evaluate the reactor operation without the addition of alkalizing agents to reduce operating costs in a biogas plant. Phase II-A was initiated with an OLR of $1.9 \text{ kg VS m}^{-3} \text{ d}^{-1}$. Under this condition, a reduction in methanogenic production was observed, resulting in a MY of $120 \pm 28 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS fed}$ and a VMP of $2,689 \pm 636 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$. Despite this decrease, the system maintained stable control over key operational parameters, including methane content (Fig. 3c), effluent pH, the IA/PA ratio, and TVA concentration (Fig. 2a-b). Furthermore, as previously discussed, VS removal in Phase II was higher than in Phase I. Therefore, despite the increased removal of organic matter, methane production specifically in Phase II-A was slightly lower than in the previous phase. These results suggest a diversion of electron flow toward cellular synthesis during this phase of operation.

Given the feasibility of the process, the OLR was progressively increased, reaching $3.0 \text{ kg VS m}^{-3} \text{ d}^{-1}$ in Phase II-B and $4.1 \text{ kg VS m}^{-3} \text{ d}^{-1}$ in Phase II-C. This gradual increase in the organic load increased methane production, reaching mean methane flow rates of $1,315 \pm 185$ and $1,550 \pm 187 \text{ NmL CH}_4 \text{ d}^{-1}$, MY of 128 ± 18 and $111 \pm 18 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS fed}$, VMP of $4,472 \pm 629$ and $5,269 \pm 637 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$, and CH_4 concentrations of $52 \pm 4\%$ and $49 \pm 4\%$, respectively. The results demonstrate that Mix 2 enabled increased OLR without requiring alkalizing agents. The higher proportion of cattle manure in Mix 2 relative to Mix 1 elevated the feed pH, confirming the effective buffering capacity of cattle manure.

These results are corroborated by different experimental results. Zhang et al. [46] evaluated the anaerobic co-digestion of food waste and cattle manure. Under semi-continuous operation, at an OLR of $10 \text{ g VS L}^{-1} \text{ d}^{-1}$, the methane production reached $317 \text{ mL CH}_4 \text{ g}^{-1} \text{ VS}$. The authors suggested that the addition of cattle manure enhanced the buffering capacity, enabling operation at high OLRs without the need for pH control. Santi Caraça et al. [47] investigated the co-digestion of vinasse with dairy cattle wastewater (DCW) in two upflow anaerobic sludge blanket (UASB) reactors arranged in series. The improvement in the nutritional profile of the vinasse due to co-digestion with DCW, combined with effluent recirculation, resulted in a VMP of $451 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$. Moraes et al. [48] evaluated the anaerobic digestion of sugar beet vinasse in CSTRs supplemented either with lime fertilizer or with 3% cattle manure. In both reactors, the C/N ratio was adjusted by the addition of straw. Biogas production reached $235.7 \pm 32.2 \text{ mL CH}_4 \text{ g}^{-1} \text{ VS}$ in the reactor supplemented with fertilizer and $265.2 \pm 26.8 \text{ mL CH}_4 \text{ g}^{-1} \text{ VS}$ in the reactor supplemented with manure under steady-state conditions. The authors suggested that anaerobic digestion was more stable when cattle manure was added to the vinasse.

In Phase III, the aim was to simulate the sugarcane off-season period, characterized by the unavailability of vinasse as a substrate for anaerobic digestion. Thus, the reactor was fed with Mix 3, composed of blood water and cattle manure in a 1:1 (v/m) ratio, with the addition of 3% glycerol (Table 2). The inclusion of glycerol was intended to compensate for the low organic content of the blood water. However, the results indicated that this formulation compromised the stability of the system, resulting in a significant reduction in methane production.

In Phase III-A, with an OLR of $5.2 \text{ kg VS m}^{-3} \text{ d}^{-1}$, the mean values observed were $867 \pm 289 \text{ NmL CH}_4 \text{ d}^{-1}$ for methane flow; $49.0 \pm 16.3 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS fed}$ for MY; $2,949 \pm 981 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$ for VMP; and $44.9 \pm 13.7\%$ for CH_4 concentration. Given the unsatisfactory performance, the organic load was reduced to $2.2 \text{ kg VS m}^{-3} \text{ d}^{-1}$ in Phase III-B, however, this strategy did not result in an improvement in reactor performance. The mean values of methane flow, MY and VMP reduced to

81.8 ± 121.2 NmL CH₄ d⁻¹, 10.8 ± 16.0 NmL CH₄ g⁻¹ VS fed and 278 ± 412 NmL CH₄ d⁻¹ L⁻¹, respectively, while methane concentration dropped to 16.0 ± 7.5%. Additionally, at this stage, the presence of residual hydrogen (1.0 ± 1.3%) was detected for the first time, which matches the TVA accumulation (Fig. 2b), followed by the impairment of the activity of methanogenic archaea. These findings corroborate the hypothesis presented in the preceding discussion, which suggested that glycerol, when present in inappropriate proportions, exerts a detrimental effect on process performance. This effect stems from its high biodegradability, which, when present in elevated concentrations, leads to the accumulation of intermediate acids. Therefore, the conditions imposed on the system were not adequate for its stability, evidencing the need for operational adjustments to prevent reactor collapse.

Fountoulakis et al. [39] evaluated the addition of crude glycerol to anaerobic digesters treating sewage sludge from water treatment plants under mesophilic conditions (35 °C). The authors reported that glycerol addition may enhance biogas production, provided it does not exceed a threshold concentration of 1% (v/v) in the feed. Concentrations above this limit were associated with imbalances in the AD process. The reactor operating with sewage sludge produced 1,106 ± 36 mL CH₄ d⁻¹ before glycerol addition and 2,353 ± 94 mL CH₄ d⁻¹ following the incorporation of 1% (v/v) glycerol into the feed. Ferreira et al. [41] also investigated the anaerobic co-digestion of sewage sludge with purified glycerol. In the first stage, higher proportions of glycerol (5–20% v/v) were added to the sludge, resulting in a low specific methane production, ranging from 0.1 to 0.4 mL CH₄ g⁻¹ VS applied. In the second stage, co-digestion with a lower proportion (1% v/v) of purified glycerol was tested; however, the methane production remained limited, reaching 10 mL CH₄ g⁻¹ VS applied.

In Phase IV, vinasse was partially reintroduced into the system, composing the mixture fed to the reactor together with blood water and bovine manure, in the proportion 1:1:1 (v/v/m), characterizing Mix 4 (Table 2). In order to prevent abrupt variations in reactor stability, the organic load applied was 2.3 kg VS m⁻³ d⁻¹. The results of this phase indicated a trend of system recovery, with a gradual increase in methane production. The mean values observed were 44 ± 38 NmL CH₄ g⁻¹ VS fed for MY, 1,191 ± 1,024 NmL CH₄ d⁻¹ L⁻¹ for VMP and 31 ± 18% for CH₄ concentration. These data suggest that the reintroduction of vinasse contributed to the resumption of methanogenic activity.

In Phase V, the proportion of vinasse in the mixture was reduced, and the reactor feed returned to include 3% glycerol. However, as previously observed in Phase III, the addition of glycerol possibly triggered inhibitory effects in the system. The results demonstrated a further reduction in methane yield, accompanied by signs of operational instability. Given this scenario, and to prevent a reactor collapse, the transition to Phase VI was implemented quickly, seeking to reestablish process stability.

In Phase VI, the reactor feed consisted of a mixture of vinasse, blood water, and cattle manure in a ratio of 1:2:2 (v/v/m) (Table 2). The main objective of this phase was to recover the system, which had experienced instability in the previous stages, and to continue the transition to the off-season period by reducing the proportion of vinasse in the feed. Although methane production was not substantial, this phase was essential for system readaptation and the pursuit of operational stability.

After 25 days, Phase VII began, designed to simulate the sugarcane off-season conditions, during which the reactor was operated without the addition of vinasse. In this phase, the feed consisted of a mixture of blood water and cattle manure, in a ratio of 1:2 (v/m) (Table 2), with an OLR of 3.5 kg VS $\text{m}^{-3} \text{d}^{-1}$. As a result, the system presented a mean methane flow rate of $885 \pm 504 \text{ NmL CH}_4 \text{ d}^{-1}$, an MY of $73 \pm 41 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS fed}$, and a VMP of $3,008 \pm 1,713 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$.

This phase was operated for 27 days, and therefore, it is believed that the mean results obtained could be further optimized, as increases in methane flow, MY, and VMP were observed throughout the period (Fig. 3a-b). The mean values for the last three data points of this phase were $1,254 \pm 173 \text{ NmL CH}_4 \text{ d}^{-1}$ for methane flow, $103 \pm 14 \text{ NmL CH}_4 \text{ g}^{-1} \text{ VS fed}$ for MY, and $4,264 \pm 587 \text{ NmL CH}_4 \text{ d}^{-1} \text{ L}^{-1}$ for VMP. In other words, the phase showed signs of stability and satisfactory performance for simulating the sugarcane off-season, operating without the addition of vinasse.

3.3 Energy potential estimation

The energy potential of the s-CSTR system was estimated based on the best energy performances obtained during the representative season periods (Phases I-D and II-B) and off-season (Phase VII) of sugarcane (please refer to Section 2.5 for further details). It should be noted that the analysis considered only the theoretical energy potential of methane ($\text{LHV} = 35.8 \text{ MJ m}^{-3} \text{ CH}_4$), disregarding its conversion into useful energy through systems such as generators, reciprocating engines, boilers, or turbines. Therefore, the losses associated with these processes were not included.

In Phase I-D, which simulated the season period, the system was fed with Mix 1, composed of vinasse and cattle manure in a 5:1 ratio, under an organic loading rate of $1.7 \text{ kg VS m}^{-3} \text{d}^{-1}$ (Tables 2 and 3). Under this condition, the estimated energy potential reached 299 MJ m^{-3} of mixture. Subsequently, Phase II-B, which also represented harvest conditions, using Mix 2, composed of vinasse and cattle manure in equal proportions (1:1) and an OLR of $3.0 \text{ kg VS m}^{-3} \text{d}^{-1}$ (Tables 2 and 3), resulted in an energy estimate of 336 MJ m^{-3} of mixture. Finally, Phase VII simulated the sugarcane off-season period, operating with Mix 7, consisting of blood water and cattle manure in a 1:2 ratio, under an

organic load of 3.5 kg VS m⁻³ d⁻¹ (Tables 2 and 3). At this stage, the estimated energy potential was 211 MJ m⁻³ of mixture. Table 4 summarizes the conditions adopted for calculating the energy estimate.

Table 4: Description of the scenarios adopted for calculating the energy estimate

SCENARIO 1	Season	Off-Season
Period	7.5 months	4.5 months
Feed	Mix 1	Mix 7
Composition	Vinasse + Cattle manure - 1:5 (v/m)	Blood water + Cattle manure - 1:2 (v/m)
Methane Yield (MJ m ⁻³)	299.4	211.2
	Vinasse	Blood water
Amount (thousand m ³ /period)	450	81
Amount (thousand m ³ /month)	60	18
	Cattle manure	Cattle manure
Amount (thousand tonnes/month)	90	162
Amount (thousand tonnes/month)	12	36
SCENARIO 2	Season	Off-Season
Period	7.5 months	4.5 months
Feed	Mix 2	Mix 7
Composition	Vinasse + Cattle manure - 1:1 (v/m)	Blood water + Cattle manure - 1:2 (v/m)
Methane Yield (MJ m ⁻³)	336.4	211.2
	Vinasse	Blood water
Amount (thousand m ³ /period)	225	81
Amount (thousand m ³ /month)	30	18
	Cattle manure	Cattle manure
Amount (thousand tonnes ³ /period)	225	162
Amount (thousand tonnes/month)	30	36

The assessment of the annual energy potential of the different co-digestion scenarios indicates that energy generation in Scenario 1 reaches approximately 59,169 MWh, while in Scenario 2, the estimated value is 56,298 MWh, representing a 5.1% higher energy yield. The main distinction between the scenarios concerns the season period: in Scenario 1, the vinasse was fully treated, requiring less manure addition, which may require the use of alkalizing agents to optimize process performance. In Scenario 2, the option was to treat only half of the available vinasse, preserving the existing infrastructure and allowing a more balanced comparison between the systems. In this case,

the 1:1 mixing ratio between vinasse and manure resulted in greater manure use, allowing for the elimination of alkalizing agents due to the buffering properties of livestock waste. However, from an environmental perspective, the other unprocessed half of vinasse would keep the bottlenecks associated with potential negative impacts on soil and water resources [7–9,49].

Considering that the estimates were based on a medium- to small-scale sugarcane mill, extrapolating the results to the state scale highlights the significant energy potential associated with the integration of vinasse and livestock waste. It should be noted, however, that the scenarios presented are indicative in nature, providing an overview of the system based on the energy potentials obtained. In an integrated approach, it is essential to consider the actual availability of substrates, as well as operational feasibility and technological and regional conditions that may influence the performance and applicability of the process on a large scale.

The biogas generated from the anaerobic co-digestion of vinasse and livestock waste offers versatile energy applications. Beyond conventional electricity generation through cogeneration systems, the biogas can be upgraded to biomethane for use as vehicle fuel or injection into natural gas pipelines. More innovative pathways include hydrogen production through biogas reforming. The choice of application should consider regional infrastructure, market conditions, and economic viability to maximize both environmental and financial benefits.

4. CONCLUSIONS

This study demonstrates that integrating agro-industrial residues optimizes biogas production through anaerobic co-digestion. Operating a semi-continuous stirred tank reactor (s-CSTR) with a mix of vinasse and cattle manure achieved process stability and high methane yields, with increased manure proportions eliminating the need for alkalizing agents. However, partial or complete vinasse substitution with blood water or glycerol compromised system stability, which caused metabolic intermediate accumulation and process inhibition. Recovery phases involving vinasse reintroduction and pH adjustment effectively restored operational performance. Notably, the final experimental phase demonstrated that substituting vinasse with slaughterhouse wastewater and cattle manure sustained biogas production at reduced volumes, confirming the feasibility of year-round reactor operation, including during sugarcane off-season. These findings establish co-digestion of livestock and sugarcane sector residues as a viable strategy for maximizing waste utilization, optimizing existing

infrastructure, and ensuring continuous biogas production throughout the year, contributing to energy-environmental sustainability in agro-industrial regions.

Funding

This work was supported by Geo Biogas & Carbon, a Brazilian company, and by the Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) [Grant Number 2017/11523-5].

Acknowledgements

The authors would like to acknowledge the support of Geo Biogas & Carbon (Brazilian company); the Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) [Grant Numbers 2017/11523-5 and 2024/01112-1]; the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) [310412/2022-6 and 405692/2022-6]; and the Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG) [APQ 05218-23 and APQ 00544-23]. The authors also acknowledge the support of the Biotechnology Laboratory for Bioenergy and Environment at the School of Agricultural Engineering (FEAGRI) and the Center for Energy Planning (NIPE), both at the University of Campinas (UNICAMP).

Author contributions:

Ana Beatriz Soares Aguiar: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Writing – original draft; Writing – review and editing; **Renata Piacentini Rodriguez:** Conceptualization; Formal analysis; Investigation; Methodology; Project administration; Supervision; Validation; Visualization; Writing – review and editing; **Lucas Tadeu Fuess:** Formal analysis; Investigation; Validation; Visualization; Writing – review and editing; **Jens Bo Holm-Nielsen:** Formal analysis; Investigation; Validation; Visualization; Writing – review and editing; **Bruna de Souza Moraes:** Conceptualization; Formal analysis; Funding Acquisition; Investigation; Methodology; Project administration; Resources; Supervision; Validation; Visualization; Writing – review and editing

REFERENCES

- [1] Zarkadas IS, Sofikiti AS, Voudrias EA, Pilidis GA. Thermophilic anaerobic digestion of pasteurised food wastes and dairy cattle manure in batch and large volume laboratory digesters: Focussing on mixing ratios. *Renew Energy* 2015;80:432–40. <https://doi.org/10.1016/j.renene.2015.02.015>.
- [2] España-Gamboa E, Mijangos-Cortes J, Barahona-Perez L, Dominguez-Maldonado J, Hernández-Zarate G, Alzate-Gaviria L. Vinasses: Characterization and treatments. *Waste Management and Research* 2011;29:1235–50. <https://doi.org/10.1177/0734242X10387313>.
- [3] Romanholo Ferreira LF, Aguiar MM, Messias TG, Pompeu GB, Queijeiro Lopez AM, Silva DP, et al. Evaluation of sugar-cane vinasse treated with *Pleurotus sajor-caju* utilizing aquatic organisms as toxicological indicators. *Ecotoxicol Environ Saf* 2011;74:132–7. <https://doi.org/10.1016/j.ecoenv.2010.08.042>.
- [4] Parsaei M, Kiani Deh Kiani M, Karimi K. A review of biogas production from sugarcane vinasse. *Biomass Bioenergy* 2019;122:117–25. <https://doi.org/10.1016/j.biombioe.2019.01.034>.
- [5] Calegari RP. Produção de biogás a partir de vinhaça concentrada. Escola Superior de Agricultura “Luiz de Queiroz” - USP, 2017.
- [6] Moraes BS, Zaiat M, Bonomi A. Anaerobic digestion of vinasse from sugarcane ethanol production in Brazil: Challenges and perspectives. *Renewable and Sustainable Energy Reviews* 2015;44:888–903. <https://doi.org/10.1016/j.rser.2015.01.023>.
- [7] Madrid L, Diaz-Barrientos E. Release of metals from homogeneous soil columns by wastewater from an agricultural industry. vol. 101. 1998.
- [8] Oliveira BG de, Carvalho JLN, Cerri CEP, Cerri CC, Feigl BJ. Soil greenhouse gas fluxes from vinasse application in Brazilian sugarcane areas. *Geoderma* 2013;200–201:77–84. <https://doi.org/10.1016/j.geoderma.2013.02.005>.
- [9] Silva MAS da, Griebeler NP, Borges LC. Use of stillage and its impact on soil properties and groundwater. *Revista Brasileira de Engenharia Agrícola e Ambiental* 2007;11:108–14.
- [10] Santana Junior AE, Duda RM, Oliveira RA de. Improving the energy balance of ethanol industry with methane production from vinasse and molasses in two-stage anaerobic reactors. *J Clean Prod* 2019;238. <https://doi.org/10.1016/j.jclepro.2019.07.052>.
- [11] Takeda PY, Oliveira CA, Dias MES, Paula CT, Borges A do V, Damianovic MHRZ. Enhancing the energetic potential of sugarcane biorefinery exchanging vinasse and glycerol in sugarcane off-season in an anaerobic reactor. *Renew Energy* 2022;195:1218–29. <https://doi.org/10.1016/j.renene.2022.06.128>.
- [12] Fuess LT, Kiyuna LSM, Ferraz ADN, Persinoti GF, Squina FM, Garcia ML, et al. Thermophilic two-phase anaerobic digestion using an innovative fixed-bed reactor for enhanced organic matter removal and bioenergy recovery from sugarcane vinasse. *Appl Energy* 2017;189:480–91. <https://doi.org/10.1016/j.apenergy.2016.12.071>.
- [13] Aquino S de, Fuess LT, Pires EC. Media arrangement impacts cell growth in anaerobic fixed-bed reactors treating sugarcane vinasse: Structured vs. randomic biomass immobilization. *Bioresour Technol* 2017;235:219–28. <https://doi.org/10.1016/j.biortech.2017.03.120>.
- [14] Borges A do V, Fuess LT, Alves I, Takeda PY, Damianovic MHRZ. Co-digesting sugarcane vinasse and distilled glycerol to enhance bioenergy generation in biofuel-producing plants. *Energy Convers Manag* 2021;250. <https://doi.org/10.1016/j.enconman.2021.114897>.
- [15] IBGE. Rebanho de Bovinos (Bois e Vacas) no Brasil. Instituto Brasileiro de Geografia e Estatística 2023. <https://www.ibge.gov.br/explica/producao-agropecuaria/bovinos/br> (accessed January 21, 2025).
- [16] Font-Palma C. Methods for the Treatment of Cattle Manure—A Review. *C — Journal of Carbon Research* 2019;5:27. <https://doi.org/10.3390/c5020027>.
- [17] Gonçalves CS. Avaliação do Potencial de Geração de Biogás a partir de Resíduos de Boviniculturas na Área Metropolitana do Porto. Dissertação de Mestrado Em Engenharia Do Ambiente - Faculdade de Engenharia Da Universidade Do Porto 2010:135.
- [18] Mathias JFCM. Manure as a resource: Livestock waste management from anaerobic digestion, opportunities and challenges for Brazil. *International Food and Agribusiness Management Review* 2014;17:87–110. <https://doi.org/10.22004/ag.econ.188711>.

- [19] Othman I, Anuar AN, Ujang Z, Rosman NH, Harun H, Chelliapan S. Livestock wastewater treatment using aerobic granular sludge. *Bioresour Technol* 2013;133:630–4. <https://doi.org/10.1016/j.biortech.2013.01.149>.
- [20] Ozturk B. Evaluation of Biogas Production Yields of Different Waste Materials. *Earth Science Research* 2012;2. <https://doi.org/10.5539/esr.v2n1p165>.
- [21] Pacheco JW. Guia técnico ambiental de abates (bovino e suíno). São Paulo: 2006.
- [22] Miao H, Wang S, Zhao M, Huang Z, Ren H, Yan Q, et al. Codigestion of Taihu blue algae with swine manure for biogas production. *Energy Convers Manag* 2014;77:643–9. <https://doi.org/10.1016/j.enconman.2013.10.025>.
- [23] Karki R, Chuenchart W, Surendra KC, Shrestha S, Raskin L, Sung S, et al. Anaerobic co-digestion: Current status and perspectives. *Bioresour Technol* 2021;330. <https://doi.org/10.1016/j.biortech.2021.125001>.
- [24] Mao C, Feng Y, Wang X, Ren G. Review on research achievements of biogas from anaerobic digestion. *Renewable and Sustainable Energy Reviews* 2015;45:540–55. <https://doi.org/10.1016/j.rser.2015.02.032>.
- [25] Usack JG, Spirito CM, Angenent LT. Continuously-stirred anaerobic digester to convert organic wastes into biogas: System setup and basic operation. *Journal of Visualized Experiments* 2012. <https://doi.org/10.3791/3978>.
- [26] APHA, AWWA, WEF. Standard Methods for Examination of Water and Wastewater. 22nd ed. Washington: American Public Health Association, American Water Works Association, Water Environment Federation; 2012.
- [27] Ripley LE, Boyle WC, Converse JC. Improved Alkalimetric Monitoring for Anaerobic Digestion of High-Strength Wastes. vol. 58. 1986.
- [28] Dilallo R, Albertson OE. Volatile Acids by Direct Titration. vol. 33. 1961.
- [29] De Paula MS, Adarme OFH, Volpi MPC, Flores-Rodriguez CI, Carazzolle MF, Mockaitis G, et al. Unveiling the biogas potential of raw Agave leaf juice: Exploring a novel biomass source. *Biomass Bioenergy* 2025;193. <https://doi.org/10.1016/j.biombioe.2024.107522>.
- [30] Formann S, Hahn A, Janke L, Stinner W, Sträuber H, Logroño W, et al. Beyond Sugar and Ethanol Production: Value Generation Opportunities Through Sugarcane Residues. *Front Energy Res* 2020;8. <https://doi.org/10.3389/fenrg.2020.579577>.
- [31] De Oliveira ACL, Resende M de O, Silva EGM, Renato N dos S, Martins MA, Sequinel R, et al. Evaluation and optimization of electricity generation through manure obtained from animal production chains in two Brazilian mesoregions. *J Clean Prod* 2021;316. <https://doi.org/10.1016/j.jclepro.2021.128270>.
- [32] Siddique MNI, Wahid ZA. Achievements and perspectives of anaerobic co-digestion: A review. *J Clean Prod* 2018;194:359–71. <https://doi.org/10.1016/j.jclepro.2018.05.155>.
- [33] Kunz A, Steinmetz RLR, Amaral AC do. Fundamentos da digestão anaeróbia, purificação do biogás, uso e tratamento do digestato. Airton Kunz, Ricardo Luis Radis Steinmetz, André Cestonaro do Amaral (editores Técnicos); 2019. <https://doi.org/10.21452/978-85-93823-01-5.2019.01>.
- [34] Probiogás. Tecnologias de digestão anaeróbia com relevância para o Brasil: substratos, digestores e uso de biogás. Brasília, DF: Cooperação Alemã para o Desenvolvimento Sustentável por meio da Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH(GIZ); 2015.
- [35] Barbosa MYU, Alves I, Del Nery V, Sakamoto IK, Pozzi E, Damianovic MHRZ. Methane production in a UASB reactor from sugarcane vinasse: shutdown or exchanging substrate for molasses during the off-season? *Journal of Water Process Engineering* 2022;47. <https://doi.org/10.1016/j.jwpe.2022.102664>.
- [36] Aslankoochi E, Rezaei MN, Vervoort Y, Courtin CM, Verstrepen KJ. Glycerol production by fermenting yeast cells is essential for optimal bread dough fermentation. *PLoS One* 2015;10. <https://doi.org/10.1371/journal.pone.0119364>.
- [37] Godoi LAG de, Camiloti PR, Bernardes AN, Sanchez BLS, Torres APR, Conceição Gomes A da, et al. Seasonal variation of the organic and inorganic composition of sugarcane vinasse: main implications for its environmental uses. *Environmental Science and Pollution Research* 2019;26:29267–82. <https://doi.org/10.1007/s11356-019-06019-8>.
- [38] Rogeri RC, Gomes KG, de Araujo MN, do Vale Borges A, Gil-Garcia C, Damianovic MHRZ, et al. Advancements in two-stage anaerobic digestion of sugarcane vinasse: overcoming drawbacks by

- exploiting the fermentative-sulfidogenic process. *Rev Environ Sci Biotechnol* 2025;24:885–909. <https://doi.org/10.1007/s11157-025-09744-4>.
- [39] Fountoulakis MS, Petousi I, Manios T. Co-digestion of sewage sludge with glycerol to boost biogas production. *Waste Management* 2010;30:1849–53. <https://doi.org/10.1016/j.wasman.2010.04.011>.
- [40] Robra S, Serpa da Cruz R, de Oliveira AM, Neto JAA, Santos J V. Generation of biogas using crude glycerin from biodiesel production as a supplement to cattle slurry. *Biomass Bioenergy* 2010;34:1330–5. <https://doi.org/10.1016/j.biombioe.2010.04.021>.
- [41] Ferreira J dos S, Volschan I, Cammarota MC. Co-digestion of sewage sludge with crude or pretreated glycerol to increase biogas production. *Environmental Science and Pollution Research* 2018;25:21811–21. <https://doi.org/10.1007/s11356-018-2260-3>.
- [42] Viana MB, Freitas A V., Leitão RC, Pinto GAS, Santaella ST. Anaerobic digestion of crude glycerol: a review. *Environmental Technology Reviews* 2012;1:81–92. <https://doi.org/10.1080/09593330.2012.692723>.
- [43] Jensen PD, Astals S, Lu Y, Devadas M, Batstone DJ. Anaerobic codigestion of sewage sludge and glycerol, focusing on process kinetics, microbial dynamics and sludge dewaterability. *Water Res* 2014;67:355–66. <https://doi.org/10.1016/j.watres.2014.09.024>.
- [44] Holm-Nielsen JB, Lomborg CJ, Oleskowicz-Popiel P, Esbensen KH. On-line near infrared monitoring of glycerol-boosted anaerobic digestion processes: Evaluation of process analytical technologies. *Biotechnol Bioeng* 2008;99:302–13. <https://doi.org/10.1002/bit.21571>.
- [45] Goembira F, Yuliarningsih R, Silvia S, Rahmadani F. The effects of crude glycerol addition on biogas production. *IOP Conf. Ser. Earth Environ. Sci.*, vol. 1306, Institute of Physics; 2024. <https://doi.org/10.1088/1755-1315/1306/1/012041>.
- [46] Zhang C, Xiao G, Peng L, Su H, Tan T. The anaerobic co-digestion of food waste and cattle manure. *Bioresour Technol* 2013;129:170–6. <https://doi.org/10.1016/j.biortech.2012.10.138>.
- [47] Santi Caraça M de, Oliveira RA de, Duda RM. Bioenergy Recovery from Anaerobic Co-Digestion of Sugarcane Vinasse and Dairy Cattle Wastewater in Two Up-Flow Anaerobic Sludge Blanket Reactors in Series. *Bioenergy Res* 2023. <https://doi.org/10.1007/s12155-023-10614-6>.
- [48] Moraes BS, Triolo JM, Lecona VP, Zaiat M, Sommer SG. Biogas production within the bioethanol production chain: Use of co-substrates for anaerobic digestion of sugar beet vinasse. *Bioresour Technol* 2015;190:227–34. <https://doi.org/10.1016/j.biortech.2015.04.089>.
- [49] Fuess LT, Garcia ML. Implications of stillage land disposal: A critical review on the impacts of fertigation. *J Environ Manage* 2014;145:210–29. <https://doi.org/10.1016/j.jenvman.2014.07.003>.